Experimental Capture/Recapture Estimation Using Census and Administrative Data

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Abstract

This report expands upon the innovation of utilizing administrative records and third-party data implemented in the 2020 Census. The 2020 Census used administrative records and third-party data in address canvassing and nonresponse followup operations. The Census Bureau also has a long history of using administrative records of births, deaths, and other information to produce Demographic Analysis coverage estimates. Since 1980, the Census Bureau has produced capture-recapture coverage estimates by conducting an independent post-enumeration survey and utilizing dual system estimation approaches. This report presents the research results of attempting to see if administrative records and third-party data could be utilized to produce capture-recapture coverage estimates. This work uses an Expectation Maximization Log Linear Modeling approach previously researched by Statistics Netherlands and Statistics New Zealand. This report documents some of the experimental results from an evaluation that was part of the 2020 Census Program for Evaluation, Experiments, and Assessments.

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1. Introduction

This research expands upon the innovation of using administrative records (AR) and third-party data for the 2020 Census Address Canvassing and Nonresponse Followup operations. The Census Bureau also has a long history of using AR of birth, deaths, and other information to produce Demographic Analysis (DA) coverage estimates[1] (Robinson et al. 1993, Robinson et al. 2002, U.S. Census Bureau 2012, Jensen et al. 2020). Since 1980, the Census Bureau has produced capture-recapture coverage estimates by conducting an independent post-enumeration survey and using Dual System Estimate (DSE) approaches. This evaluation expands this 2020 innovation by attempting to see if administrative records and third-party data could be used to produce capture-recapture coverage estimates.

Because of budget constraints, the 2020 Census Post-Enumeration Survey (PES) had limited resources to research and test improvements for the necessary housing unit and person operations leading up to the 2020 Census. These operations included the costly field and clerical matching procedures necessary for a sample-based application of the dual-system estimation. For example, the 2010 Census coverage survey had field operation costs of $15.2 million for Independent Listing, $23.7 million for Interviewing, and $14.9 million for additional person follow-up (Contreras et al. 2012, Linse and Argarin 2012, Johnson et al. 2012). These costs did not include processing system development costs. Due to the lack of research and testing, there was limited opportunity to reduce the costs and increase the efficiency of those operations for the 2020 Census PES[2].

For the 2018 End-to-End Census Test, the Post-Enumeration Survey was planned but canceled. As a result, 2020 Census PES operations were implemented for the first time during the 2020 Census using the same costly design as the 2010 Census Coverage Measurement program. Based on these developments, we conducted an evaluation to examine whether AR could be used to produce net coverage estimates in the same manner as the survey-based results without having to implement the field collection, clerical matching software development, and incurring clerical matching personnel costs. This evaluation used AR as the second system instead of the independent PES sample. The census continued to serve as the first system. This evaluation implemented a methodology

[1] Demographic Analysis program has produced limited estimates by Hispanic Origin since the 2010 Census. https://www.census.gov/programs-surveys/decennial-census/about/coverage-measurement/da.html

[2] The Post Enumeraiton Survey was called the Accuracy and Coverage Evaluation in 2000 and Census Coverage Measurement in 2010. More informaiton about the program can be found at https://www.census.gov/programs-surveys/decennial-census/about/coverage-measurement/pes.html

to produce coverage estimates for the 2020 Census and compared the results with the estimates of census coverage from the 2020 DA and Population Estimates Program[3] (PEP).

2. Background

This section provides background related to both coverage measurement and some of the usage of AR in the 2020 Census enumeration. Section 2.1 discusses and provides some historical information about the use of post-enumeration surveys to measure census coverage. Section 2.2 provides background on the research for the increased use of administrative data in the 2020 Census. Section 2.3 provides additional information about approaches applied by other statistical agencies across the world.

2.1 Post-Enumeration Surveys for the Decennial Census

The U.S. Census Bureau has a history of producing population estimates using dual-system estimation from a sample survey. Wolter (1986) gave an overview of the methodology producing a population estimate using two lists or systems and provided the assumptions that need to be met to produce valid population estimates. The assumptions included (1) being included in the census is independent of being included in the survey, (2) erroneous inclusions in each list are removed, (3) there is no matching error, and other assumptions.

For census applications, one system is the Enumeration or E system. This starts with the enumerations in the census. The second system is the Population or P system. This is an independent enumeration of the population. The Wolter (1986) paper outlined how DSEs could be produced using the Petersen model to generate valid estimates in post-strata. Synthetic estimation could then be used to produce small area estimates below the post-strata level.

Hogan and Wolter (1988) documented the 1980 Census coverage estimates. For the 1980 survey, the Current Population Survey was used as the second system. The 1980 analysis produced a series of alternative estimates based on different assumptions. Lessons learned from their application were incorporated into the design for the 1990 PES.

Hogan (1993) documented the production of population estimates for the 1990 PES. The 1990 PES used an independent listing of sampled census blocks instead of the Current Population Survey. Independent interviews were conducted at those addresses. This included collecting where the person should have been counted on Census Day. The PES matched the independently rostered people to the areas identified where they should have

[3] The Census Bureau's Population Estimates Program produces estimates of the population for the United States, states, metropolitan and micropolitan statistical areas, counties, cities, towns, as well as for Puerto Rico and its municipios. https://www.census.gov/programs-surveys/popest.html

been counted. Computer and clerical matching were used to determine the residence and match statuses of the cases. Follow-up interviews were conducted based on stringent criteria to resolve cases. The final population estimates were based on calculating the DSEs for 1,392 post-strata (later condensed to 357 post-strata).

Hogan (2003) documented the production of population estimates for the 2000 Accuracy and Coverage Evaluation (A.C.E.) of the 2000 Census. The 2000 A.C.E. was like the 1990 PES in that independent listing and interviewing on a sample basis were conducted. Also, like 1990, computer, clerical, and follow-up operations were conducted.

The initial 2000 coverage survey estimates showed a difference of about 3 million people as compared with those produced by the DA program (U.S. Census Bureau 2001). Robinson et al. (2002) documented how estimates of births, deaths, immigration, and emigration are constructed to produce population estimates. Analysis of the initial A.C.E. estimates led to two sets of revisions. The final revisions accounted for erroneous inclusions in both systems and matching error. The final estimates also included a correlation bias adjustment for the violation of independence between the captures. The main change in the estimates was due to erroneous inclusions in the census system that had not been removed for the original estimates. A major source of the erroneous inclusions was person duplication not identified as erroneous by the coverage program.

Mule (2012) documented the person coverage estimates from the Census Coverage Measurement (CCM) survey of the 2010 Census. Like the 1990 and 2000 coverage surveys, the 2010 CCM evaluated net census coverage error by using DSE to generate population estimates. New for 2010, the CCM used logistic regression modeling instead of poststratification to produce synthetic estimates of net coverage. The logistic regression modeling allowed for the reduction of correlation bias in the total population estimates without having to include unnecessary higher-order interactions as when forming poststratification cells. Like the 2000 A.C.E., the 2010 CCM included an adjustment for remaining correlation bias for some of the population estimates.

2.2 Administrative Records Research for the Decennial Census and Post-Enumeration Surveys

The Census Bureau has conducted research using AR during the decennial census enumeration. Leggieri et al. (2002) documented the AR census experiment for the 2000 Census. This was an experiment to see if AR could be used to conduct the decennial enumeration for two counties in Maryland and three counties in Colorado. The results showed potential undercoverage when solely using AR for enumeration at that time.

The 2010 Census also included multiple evaluations involving AR. Sheppard et al. (2013) described an evaluation that used AR to detect and reduce overcoverage. While not examining undercoverage, this 2010 evaluation recommended that future studies about improving census coverage with AR include a follow-up component to assess discrepancies between the census and AR. The 2010 Census Match Study (Rastogi and O'Hara, 2012) assessed the quality and coverage of several AR and third-party sources relative to the 2010 Census. Relevant to this research, Rastogi and O'Hara found that the AR and third-party sources had less coverage for harder-to-count populations.

Keller et al. (2018) documented research on how AR and third-party data were used to reduce contacts in the Nonresponse Followup operation in the 2020 Census. This work included building rosters from AR sources and determining if there was sufficient information from the AR to reduce the number of times a fieldworker attempted to visit an address to obtain a census response. Variations of this approach were implemented in the 2013, 2014, 2015, 2016, and 2018 census tests. Results from each test were used to refine and improve the methodology for the next test. Mulry et al. (2021) documented the implementation of using AR for enumeration in the 2020 Census. Our research involves similar evaluations by using AR to build household rosters independent of the concurrent census.

As part of this research, we built on previous research and suggestions using AR for census coverage estimation. Ericksen and Kadane (1986) proposed a way to use administrative lists to estimate census omissions. Zaslavsky and Wolfgang (1993) and Griffin (2014) proposed the usage of AR as the source of a third list to go with the census and a PES that would allow multiple system estimation.

2.3 Approaches Used by Other Statistical Agencies

With the increased availability of administrative data, other countries have been exploring how to use this information to produce population estimates based on two or more sources. This subsection gives an overview of the methods that inspired our evaluation, including uses of AR to create two systems that could have application to our net coverage estimation problem.

Statistics Netherlands has been researching a novel capture-recapture approach. Van der Heijden et al. (2018) documented a capture-recapture approach that used a combined data file that treats information not observed as a missing data problem. Their paper documented a log-linear modeling approach using an Expectation-Maximization (EM) algorithm. The paper outlined a novel approach where the research item of interest is captured differently by two sources. Their paper showed the example of estimating the

numbers of road accidents by cause as reported to either the police or at a hospital. In thinking ahead to having three or more sources available, van der Heijden et al. (2021) provided an application to estimating the Māori population for Statistics New Zealand.

As Wolter (1986) discussed, there are various assumptions that are required in capture-recapture to obtain unbiased estimates. Yildiz et al. (2017) discussed a methodology to estimate population estimates when the assumptions of homogeneous inclusion probabilities or overcoverage were violated. Their approach adjusted capture-recapture estimates from a loglinear model to available auxiliary control totals. The goal was to compensate for situations where administrative record sources had erroneous inclusions. For our research, we applied this adjustment approach where there may be both erroneous inclusions and omissions in the sources.

3. Methodology

This section details the methodology used in the evaluation. Since we did not have any fieldwork, the focus is on how the data was constructed and the details of the statistical methods investigated.

3.1 Estimation Methodology

This section provides the estimation methodology developed and implemented for this analysis. We start by giving an overview of the particular capture-recapture methodology that was researched. This section also documents how the census and administrative record data were processed for people and their characteristics. The section concludes by providing details on the estimation methodology used to produce the various population net coverage estimates.

This research used the population size estimation methodology developed by van der Heijden et al. (2018). They developed an approach using log-linear modeling and the Expectation-Maximization (EM) algorithm. Their paper put forth a method for two systems that accounted for whether a concept of interest was potentially measured differently by each system. The paper used the example of estimating traffic accidents by cause, using data from two reporting systems. In this section, we describe how we implemented this methodology in our study using the census and administrative data as the two systems where each separately measures race and Hispanic origin.

To compute coverage estimates, two population systems were defined, the administrative records (AR) population and the census population. We constructed the AR population by

combining various AR datasets together. To enable the linking of the varied administrative datasets, an anonymized identifier was assigned to each person record in each administrative record file. The Census Bureau's Person Identification Validation System (PVS) determined the protected identification key[4] (PIK) through a probabilistic matching algorithm between the administrative record source data and a series of reference files (Wagner and Layne, 2014). Given the PIK assignments, person records from the various administrative files, as well as the census, could then be matched through the PIKs.

The AR population comprised the union of four component AR subpopulations (described in more detail below):

(1) AR sources used in the 2020 Census to form rosters for enumeration.
(2) Bureau of Prisons (BOP) rosters of incarcerated people.
(3) People receiving benefits from a subset of states that provided their Supplemental Nutritional Assistance Program (SNAP) or Temporary Assistance for Needy Families (TANF) program data[5].
(4) People receiving benefits from Medicaid.

The first AR subpopulation constituted the bulk of the total AR population. With respect to occupied enumerations using AR in the 2020 Census, the 2020 vintage AR sources used to create 2020 AR household rosters were:

- Tax Year (TY) 2019 Internal Revenue Service (IRS) Individual Tax Returns (Form 1040).
- TY 2019 IRS Informational Returns (Form 1099).
- Indian Health Service (IHS) Patient Database.
- Centers for Medicare and Medicaid Services (CMS) Medicare Enrollment Database.
- Census Household Composition Key File.

For distinct person records found in at least one of the five federal AR sources above, we applied certain criteria to determine if the person should be included in the AR population. Using the 2020 Census Number Identification (Numident) file, we verified that the person was alive on April 1, 2020, and less than 116 years old. We also required that the person record must be assigned a Master Address File[6] ID (MAFID) so we could assign the person

[4] Protected Identification Key (PIK) is an encrypted anoonyous identifier that is unique as a Security Number or a Tax Identificaiton Number. https://www.census.gov/content/dam/Census/library/working-papers/2014/adrm/carra-wp-2014-01.pdf
[5] The states included in the same SNAP/TANF datasets that were used include Arizona, California, Hawaii, Idaho, Iowa, Kentucky, Maryland, Massachusetts, Mississippi, Montana, Nebraska, New Jersey, New York, North Carolina, North Dakota, Oregon, South Carolina, Tennessee, Utah, and Wyoming.
[6] The Master Address File (MAF) is a Census Bureau file that contains an accurate, up to date inventory of all known living quarters in the United States, Puerto Rico and associated island areas.

to a specific address rather than a geographic area for this initial work. Requiring a MAFID for a person to exist on the AR roster may create an independence issue. For areas where we have 1) AR people without MAFIDs and 2) a lower PIK rate in the census, we could have decreased coverage because of this correlation.

The IRS 1040 and 1099 sources identified people who had 1040 or 1099 returns on file with the previous tax year. Presence on the IHS or Medicare databases indicated that a person was eligible to receive services for those programs. The Household Composition Key file was the last file that we used for 2020 AR household rosters. This file connected children with their mothers and/or fathers where the linkages were able to be made. While the first four AR sources consisted of addresses and PIKs, the Household Composition Key File did not contain address information. However, that was obtained by linking to other files when needed. The record for a child on the Household Composition Key File included the names of the mother and father. Errors in the parent PIK assignment were possible. The strategy adopted for 2020 was to add the child to the AR roster (if not already present) for any of the housing units in which either the mother or the father appeared (Konicki, 2017). Hence, if the father or mother appeared on any of the first four AR sources, their child from the Household Composition Key file was added to the roster.

The second AR subpopulation entailed BOP rosters of incarcerated people in those prisons. The purpose behind including BOP rosters was to partially cover the group quarters (GQ) population. Note that BOP people were counted at the prison address, as opposed to the address with which they were associated before entering the BOP facility. This is consistent with census residence rules.

The third AR subpopulation consisted of people receiving benefits from the SNAP or TANF programs. In contrast to the first two subpopulations, SNAP and TANF programs are administered by the states and the datasets are procured through the states. As a result, we used this state-level program data from a selected set of states. The idea behind including SNAP and TANF data in this evaluation was that they are believed to include many people from groups who are historically undercounted in the census.

A fourth subpopulation consists of people who received Medicaid benefits per the 2019 Transformed Medicaid Statistical Information System (T-MSIS) database. Like the SNAP data above, the T-MSIS database provides additional records for groups who are historically undercounted in the census. However, unlike the SNAP data, which are procured from the states, the T-MSIS data are organized into a national database.

It is important to note that AR were only intended to be used to enumerate a subset of the population for the 2020 Census. That is, AR were only used 1) when they were determined to be of sufficiently high quality, and 2) when a census response was not collected through an internet, mail, or enumerator response. Consequently, given time and processing constraints, the first AR subpopulation described above was the only AR subpopulation used during the 2020 Census to enumerate cases with AR. This is relevant because those 3.20 percent of addresses that were enumerated using AR will largely have the same rosters in the census population and the AR population. Differences may occur if additional people on the AR roster where from SNAP, TANF, or T-MSIS. This might cause correlation bias in DSE. BOP data were not used in the census because GQ addresses were out-of-scope for census AR enumerations. SNAP and TANF data were not used for census AR enumerations because we did not have this information available for all states. T-MSIS data were not used for census AR enumerations because they were not available during the processing cycle.

After taking the union of the four AR subpopulations, the resulting AR household roster, the collection of PIKs found in any of the selected AR files at a given address, was unique by person and address. That is, no individuals were duplicated within a housing unit. However, the same PIK may have occurred at multiple MAFIDs. As a result, we had to unduplicate that PIK so it was at only one MAFID.

When the same PIK existed at multiple MAFIDs, a subsequent unduplication step was undertaken to ensure that the PIK was included only once in the AR population. This unduplication step was done in a hierarchical manner. First, we checked to see if the duplicated person was counted in both a GQ and a housing unit (HU). If counted in both, we kept the person record that was in a GQ. Second, we checked to see how many sources the person was associated with at that address. In general, the more AR sources, the more likely we were to keep them at that address. Third, we checked to see the address at which the person was enumerated in the 2010 Census. Generally, we kept the person record at the address that was closer to their 2010 Census residence. Lastly, if the unduplication could not be achieved as just described, we randomly picked the address at which to keep the AR person.

To define the census population for use in DSE, we took the 331 million people in the 2020 Census and removed all the census enumerations for which a PIK could not be assigned. We then linked the AR and census systems using the PIK. This left about 279 million people in the 2020 Census with a PIK. Like what was done for the AR population, we then had to unduplicate the census records so that a census PIK was unique in the census population

file. Note that we could have linked the census and AR population by other matching techniques. However, the PIK was used as the identifier that linked census persons to AR in the 2020 Census, so its use was carried over into this research.

We proceeded with unduplication of census PIKs in a hierarchical manner like what we did for the AR population. For the census duplications however, we chose the person record to retain based on the operation in which the records were enumerated. If one record was enumerated by self-response (enumeration by internet or mail) while the other record was enumerated through a field operation such as Nonresponse Followup, we took the person record associated with the self-response enumeration. This decision assumed that the self-response was closer to Census Day and was more likely to have been completed by someone associated with the household. In the scenario that the duplication was not reconciled by examining the operations in which the duplicates were enumerated, we then performed a random draw.

Table 1 shows the breakdown of counts after the AR and census populations were determined. Table 1 shows that the intersection between the AR and census populations included 261.5 million people. The number of people included in the AR population who were not in the census population, with the reductions noted for unduplication and to exclude people without PIKs, was 52.9 million. This was substantially larger than the number of people included in the census but not included in the AR population (11.1 million). The point of this research is to develop counts for subgroups within the empty boxes in Table 1. Doing so will allow for estimates of coverage of the 2020 Census for those various subgroups.

Table 1: Observed Cells of the AR Population and Census Populations

	In AR	Not in AR	Total
In 2020 Census	261,500,000	11,100,000	272,600,000
Not In 2020 Census	52,920,000		
Total	314,400,000		325,500,000

Source: U.S. Census Bureau, 2020 Adrec DSE and 2020 Census Edited File

A goal for this research was to develop estimates of census coverage by various demographic characteristics. Consequently, the next step was to impute for missing demographic characteristics in person records in both populations, regardless of whether they are in both systems. For this analysis, we assigned sex, age, race, and Hispanic origin when they were missing. With respect to sex and age, for PIKs in both the census

population and AR population, we used the Census Numident[7] values when available, and then the reported or imputed values from the census. Sex and age imputations for people only in the census used the reported or imputed values from the census, and sex and age imputations for people only in the AR population used the values from the Census Numident.

For race and Hispanic origin, the census population again used the reported or imputed values from the census. For the AR population, we developed an assignment hierarchy to assign race and Hispanic origin since there was not a single AR source that covered the population for race and Hispanic origin like the Census Numident was used to impute sex and age. Note that, we did not use the 2020 Census race and Hispanic origin for the AR people who matched to the 2020 Census to keep the assignments independent. Table 2 shows the hierarchy used to assign race and Hispanic origin for the AR population.

Table 2: Race and Hispanic Origin Assignment Hierarchy for AR Population

Characteristic	Hierarchy
Race	1. Center for Administrative Records Research and Application (CARRA) Best Race and Hispanic Origin File – Multiracial Value 2. 2010 Census Report 3. Census Numident Place of Birth 4. Center for Administrative Records Research and Application (CARRA) Best Race and Hispanic Origin File – Non-Multirace Value 5. Medicaid Reports 6. SNAP Race Reports from state files
Hispanic Origin	1. 2010 Census Report 2. Census Numident Place of Birth 3. CARRA Best Race and Hispanic Origin File Hispanic Origin Value 4. Medicaid Reports 5. SNAP Hispanic Origin Reports from state files

With respect to census race imputation for the AR population, the first step was to assign any Multiracial values from the CARRA Best Race and Hispanic origin file[8]. After that, the general path for imputing both race and Hispanic origin was to use any race or Hispanic origin reports from the 2010 Census. We then used the Census Numident Place of Birth to

[7] Census Numident is a processed file with updates from the SSA transaction records that includes person-level informaiton including births, place of birth, country of origin and deaths. Finlay and Genadek (2021) provide more information about this data source. https://pmc.ncbi.nlm.nih.gov/articles/PMC8495642/pdf/AJPH.2021.306217.pdf

[8] CARRA Best Race and Hispanic origin file provides characteristic values based on research of business rules using several sources. https://www.census.gov/content/dam/Census/library/working-papers/2015/adrm/carra-wp-2015-08.pdf

assign a race or Hispanic origin value. For people who did not provide race or Hispanic origin or both in the 2010 Census, those whose previous ACS responses indicated a place of birth outside the United States were associated with a race or Hispanic origin code consistent with the assignment in the 2020 Census Edit and Imputation processing. Next, we used the (single) race or Hispanic origin value from the CARRA Best Race and Hispanic Origin File. Ennis et al. (2015) explains how Hispanic origin and race were assigned to people in AR data. We call this the AR Best Hispanic Origin and Race file. Note that this AR data file does not have the detailed Hispanic information (e.g., Mexican, Cuban, etc.). Next, we used the Hispanic origin or race reports from the T-MSIS dataset. Last, we used SNAP reported race from the various state files.

For the people where race or Hispanic origin were not present in any previous census or AR source, we developed separate imputation models so that all AR values could be filled for race and Hispanic origin. With respect to race, a multinomial logistic model was developed with the dependent variable being the seven-category race status (White, Black, American Indian/Alaska Native (AIAN), Asian, Native Hawaiian or Pacific Islander (NHPI), Some Other Race (SOR), or more than one race (MT1)). With respect to Hispanic origin, a binary logistic model was developed with the dependent variable being the Hispanic origin status. For both models, the independent variables included U.S. birth state, age groups, presence of people in the HU of different race groups, and block-group level percentages of the poverty rate, Hispanic rate, Asian rate, vacancy rate, renter rate, and Black rate obtained from the Planning Database of the American Community Survey[9]. Table 3 shows the use rate of each of the sources for the 314.4 million total cases in the AR population. Note that some AR sources had only Hispanic origin or only race reported, which is why the percentages are different when comparing the columns.

Table 3: Race and Hispanic Origin Use Rates of AR sources for AR population

AR Assignment Source	Race	Hispanic Origin
Best Race and Hispanic Origin File – Multiracial Report	2%	N/A
2010 Census	68%	70%
Census Numident Place of Birth	6%	2%
Best Race and Hispanic Origin File	10%	9%
Medicaid	3%	1%
SNAP	1%	1%
Imputation	10%	17%

Source: U.S. Census Bureau, 2020 Adrec DSE, 314.4 million total cases in AR population

[9] The Census Bureau's Planning Database (PDB) contains select operational, housing, demographic, and socio-economic statistics from the Decennial Census and the American Community Survey (ACS) 5-year files. https://www.census.gov/topics/research/guidance/planning-databases.html

The data creation produced a person-level file with the system (AR or census) and race/Hispanic origin information from the census and AR. We now document some of the additional variables that were assigned to the people that were used in the loglinear models explained in the next subsection.

The first additional variable was a 2010 PIK indicator. For people who were age 10 or older, we assigned a binary indicator if the PIK was found at any address in the 2010 Census file. Social Security or tax identification numbers were not collected in the census. For a PIK to have been assigned to a 2010 Census person record, the census enumeration needed to have name and date of birth information that could be assigned a PIK by the Person Validation System. For children who are age 9 and younger, they were not alive in 2010 so they could not have been counted in the 2010 Census. For these children, we made a four-level indicator based on whether the mother and/or father were found in the 2010 Census. For each 2020 child, we used the Census Household Composition Key file[10] to attempt to obtain the PIK for the mother and/or father if available. We then proceeded to see if the mother and/or father had their PIK assigned to a 2010 Census enumeration. The four levels reflect the binary outcomes of finding the mother and/or the father in the 2010 Census. This 2010 PIK indicator was identified to help reduce the heterogeneity in our coverage estimates.

In our evaluation, we ran several models to develop different population estimates. For some of the models, we attempted to account whether the person was in an area that had a higher concentration of potentially harder-to-count populations. The first variable was a three-level grouping variable based on the ACS 5-year estimate of the Hispanic concentration in the tract. The three levels reflected whether the address was in a tract that had the following concentrations of Hispanic population: a) 0 to 9 percent, b) 10 to less than 50 percent or c) 50 percent or more. A person was assigned to one of these three groups by using the tract associated with the selected MAFID (Master Address File identification number) address if the person was found in the census. If the person was not found in the census, then the assignment was made based on the tract associated with the selected MAFID address of the administrative record. The second variable was another three-level grouping variable that was assigned in a similar manner but based on the ACS 5-year estimate of the non-Hispanic Black alone population in the tract. In this application,

[10] The Census Household Composition Key is produced using SSA data and previous census informaiton linking children 18 years and younger with their parents. https://www2.census.gov/programs-surveys/decennial/2020/program-management/planning-docs/administrative-data-used-in-the-2020-census.pdf

we only allowed race and Hispanic origin to have been recorded differently in each system. We wonder if possible future applications of this work could also allow the local address covariate to be observed differently in each system.

The EM log-linear modeling methodology was run separately for each of 16 age/sex groups listed in Table 4. For each of these age/sex groups, the three models listed in Appendix A were run. These models used the 2020 Census race that includes some other race as a response. Running multiple models was one way of showing uncertainty in our resulting population estimates as compared to the bootstrap method suggested in the van der Heijden et al. (2018) paper.

Table 4: Age Sex Groups in this Analysis

0-to-4 Males	0-to-4 Females
5-to-9 Males	5-to-9 Females
10-to-17 Males	10-to-17 Females
18-to-24 Males	18-to-24 Females
25-to-29 Males	25-to-29 Females
30-to-49 Males	30-to-49 Females
50-to-64 Males	50-to-64 Females
65+ Males	65+ Females

Following the advice laid out in van der Heijden et al. (2018), we generated race and Hispanic origin estimates by summing over the cases within the race/origin group defined by the system we believed captured this more accurately. Since the 2020 Census had updates to the race and Hispanic origin questions and response options as well as enhancements to the processing of write-in fields during postprocessing, we generated estimates for groups defined by the 2020 Census race/origin categories by summing people within these categories. The resulting tables were used in the log-linear modeling.

Analysis showed that the EM log-linear modeling results appeared to be underestimates of the population when compared to DA totals for several of the age/sex groups. This general underestimation is related to several possible factors including a) violation of independence between the sources, b) heterogeneity in the capture probabilities not accounted for by the model and c) administrative record sources not covering the full population. To mitigate this potential bias, we implemented a bias adjustment to account for a violation of assumptions.

The equation for the bias adjustment for the violation of assumptions for each age and sex group separately for the low, middle, and high DA estimate is:

$$Bias\ Adj\ Adrec\ DSE_i = Adrec\ DSE_i \times \frac{DA}{\sum_i Adrec\ DSE_i}$$

where i indexes the 14 race and Hispanic origin groups including some other race.

The EM log-linear modeling results were adjusted by the two-group method to available DA age/sex totals. The DA program released three sets of results for a low, middle, and high series based on different assumptions. To help reflect uncertainty in our estimates, the EM log-linear results for each model were adjusted to the low, middle, and high estimates from DA for that age/sex group. This allowed three bias adjusted estimates for each model to be produced. The two-group methods adjusted the EM log-linear modeling estimate by the necessary factor so that the results population estimates were equal to the low, middle, or high DA estimate. Table 5 shows, for each age/sex group, the DA middle series estimates, which are followed by the four 2 x 2 table cells analogous to those of Table 1 but expressed as percentages of the DA estimates and then followed by the bias adjustments also expressed as percentages of the DA estimates. The table shows that the bias adjustment for 18- to 24-year-old females was about 0 percent, while 30- to 49-year-old males required a bias adjustment of 2.5 percent. One concern about larger bias adjustments like the 2.5 percent for the 30- to 49-year-old males is that the resulting population estimates after this adjustment may not adequately account for these biases that impact the usefulness of the resulting estimates.

Table 5: Violation of Assumptions Bias Adjustment for National Model

			Percent				
Age	Sex	DA Middle	%Both Census and Adrec	%Census Only	% Adrec Only	% Not Captured by Either	%Bias Adj
0 to 4	Male	9,948,000	70.6%	4.8%	20.2%	2.5%	1.9%
0 to 4	Female	9,510,000	70.6%	4.8%	20.3%	2.5%	1.8%
5 to 9	Male	10,460,000	77.6%	1.5%	18.8%	0.5%	1.6%
5 to 9	Female	9,974,000	77.7%	1.5%	18.9%	0.5%	1.4%
10 to 17	Male	17,790,000	79.6%	1.4%	17.0%	0.4%	1.7%
10 to 17	Female	16,980,000	79.8%	1.3%	17.0%	0.4%	1.4%
18 to 24	Male	15,420,000	72.2%	3.6%	22.6%	1.3%	0.2%
18 to 24	Female	14,800,000	74.5%	2.8%	21.6%	1.0%	0.1%
25 to 29	Male	11,660,000	70.6%	3.2%	24.7%	1.4%	0.0%
25 to 29	Female	11,310,000	75.6%	2.3%	21.2%	0.7%	0.2%
30 to 49	Male	43,460,000	73.4%	3.7%	19.1%	1.2%	2.6%
30 to 49	Female	42,890,000	78.7%	3.3%	15.5%	0.8%	1.7%
50 to 64	Male	31,580,000	79.9%	4.2%	13.5%	1.0%	1.4%
50 to 64	Female	32,530,000	83.5%	4.6%	10.4%	0.7%	0.8%
65+	Male	24,470,000	86.2%	3.6%	8.7%	0.5%	0.9%
65+	Female	29,840,000	87.4%	3.5%	8.4%	0.5%	0.2%

Source: U.S. Census Bureau, 2020 Census Edited File, 2020 Demographic Analysis and 2020 Adrec DSE

Our initial data creation and modeling produced results for 14 combined race and Hispanic origin groups. This was a cross classification of seven race categories that included some

other race as was collected in the 2020 Census. The PEP and the DA use administrative record data where the race information available does not include Some Other Race. Thus, these two produced results for race groups that did not include some other race. The PEP produced results for six race categories. This was each race alone (White alone, Black alone, AIAN alone, Asian alone and NHPI alone) and then a sixth multiracial group. When crossed with Hispanic origin, this resulted in 12 groups of race and Hispanic origin information. The DA program produced one set of results for the two groups of Black alone and non-Black alone[11]. Our evaluation used synthetic estimation methodology to allocate the 14 groups that include some other race to the 12 groups that do not. See Jensen and Johnson (2021) for more information about comparing race data to demographic benchmarks.

The synthetic allocation formula for the administrative record DSE for modified race categories is given below. These bias-adjusted population estimates for the 12 modified race groups without some other race could be summarized to produce population estimates for aggregated domains. This paper uses the modified race values produced on the Census Edited File.

$$Bias\ Adj\ Adrec\ DSE_j = \sum_{i=1\ to 14\ rh\ groupings} \left(Bias\ Adj\ Adrec\ DSE_i \times \frac{Census_{j\,|\,i}}{\sum_{j\,|\,i} Census} \right)$$

Where j indexes the 12 modified race/Hispanic origin groups without some other race, i indexes the 14 race/Hispanic origin groups with some other race, and $Census_{j\,|\,i}$ is the census count for jth modified race group within the census 14 race groups.

Our evaluation reports the net census coverage error from population estimates calculated using our method, DA, and the PEP. Table 6 shows the seven possible methods and models that produced the estimates that will be shown as net coverage error expressed as a percentage in various parts of the results section. For each of the estimates, the net coverage error estimates are calculated as follows. A positive estimate would indicate an overcount based on that result while a negative estimate would indicate an undercount.

$$Net\ Census\ Coverage\ Error\ as\ a\ Percentage = \frac{Census - Estimate}{Estimate} \times 100$$

[11] DA also produced results for Black alone-or-in-combination and non-Black alone-or-in-combination.

The van der Heijden (2018) paper had a bootstrap approach to estimate the uncertainty of their EM log-linear population estimates. Based on the millions of records in this analysis, the bootstrap methodology produced uncertainty measures that were very, very small so they were not included in this paper. For this paper, we show the results for estimates from three models and the three bias adjustments to the low, middle, and high DA age/sex results to show some measure of uncertainty. One area for future research of this application could be the development of statistical methods to allow better quantification of the uncertainty in the population estimates.

Table 6: Methods and Models Details in Study

<table>
<tr><td>Methods and Model label</td><td colspan="2">Methods and Model Details</td></tr>
<tr><td>ADREC DSE Model 1</td><td rowspan="4">This is the implementation of the EM log-linear methodology in our study.

The point estimate is the EM log-linear results adjusted to the DA middle estimate.

This analysis attempts to reflect uncertainty in the EM log-linear results by showing the range of results obtained with adjustment to either the DA low or DA high estimates.

The next column highlights differences for each model implementation.</td><td>National model in Appendix A.</td></tr>
<tr><td>ADREC DSE Model 2</td><td>National model with additional ACS 5-year tract Hispanic concentration group covariates and interactions in Appendix A.</td></tr>
<tr><td>ADREC DSE Model 3</td><td>National model with additional ACS 5-year tract non-Hispanic Black alone concentration group covariates in Appendix A.</td></tr>
<tr><td>Population Estimates</td><td colspan="2">These use the population estimates produced in 2021 for an April 1, 2020, reference date.</td></tr>
<tr><td>DA Low</td><td rowspan="3">DA estimates released in December 2020.

Please see Jensen et al. (2020) for more information.</td><td>Using different assumptions to produce the low series.</td></tr>
<tr><td>DA Middle</td><td>Using different assumptions to produce the middle series.</td></tr>
<tr><td>DA High</td><td>Using different assumptions to produce the high series.</td></tr>
</table>

3.2 Contrasting the Administrative Record and Survey-Based Approaches

This section will contrast the differences between the AR and survey-based approaches to conducting capture-recapture analysis of census populations. We start by examining how each approach corresponds to different assumptions for capture-recapture analysis that were laid out in Wolter's 1986 JASA article. The implementation of survey-based methods that adhere to these assumptions has been the basis of the U. S. Census Bureau's DSE approach since 1990. The section will then conclude by highlighting advantages and disadvantages for administrative record versus survey-based approaches in relation to a) use of sampling, b) the relevance and quality of the data and c) the matching implications especially related to the search area used.

Wolter (1986) laid out possible models and implementations of capture-recapture applications for census coverage measurement. The list/time and heterogeneity variation (Mth) model that was first proposed by Sekar and Deming (1949) allows for both time variation and heterogeneity in the capture probabilities. This approach has been the foundation of the survey-based capture-recapture methods that the U.S. Census Bureau has used since 1990. This subsection lays out some key assumptions for implementing the Mth model that are handled differently by the EM log-linear approach in our paper.

Lack of spurious events is the first assumption, which is that erroneous inclusions in either list are removed. This includes duplicates and people who died before Census Day or were born after Census Day. The efforts made to justify this assumption differ for the two approaches. For the census, determination of erroneous inclusions is typically based on information obtained from personal interviews. This survey-based approach also is able to conduct additional follow-up interviews to collect more information to help with this determination. This information obtained for the sample areas is then used to form weighted estimates of erroneous enumeration rates so these can be accounted for in the DSE. Our implementation of the log-linear modeling EM approach used birth and death records to check for and remove individual records of people who died before or were born after Census Day. We also used matching by PIKs and business rules to separately unduplicate the census and AR universes.

The efforts made to remove erroneous inclusions from the complete census and AR systems were, of necessity, more limited than the efforts made on a sample basis for the PES. We did not obtain any information about the administrative records cases related to correctness of their inclusions from personal interviews. We would thus expect some erroneous inclusions found in the PES would not be found in the AR approach. Geographic

correctness of enumerations, which matters for subnational estimates, though not for national estimates, was not considered.

Matching and Nonresponse involve a second set of assumptions. The matching assumption is that records for a person on both lists can be linked accurately. Wolter (1986) defines the Nonresponse assumption in his paper as records having sufficient information for the definition to be applied to the person so that accurate matching can be made. This can allow missing at random assumptions to be used to compensate for nonresponse. For Mth, the survey-based application applies a sufficient information definition that requires a reported name and some minimal amount of reported characteristics. Since the survey-based approach involves an additional contact to interview the population, this strategy has timing implications. The independent interview happens after the census enumeration is essentially completed. For matching, the survey-based approach must decide how the interview will handle the people who have moved since Census Day. The survey-based approach has implemented procedures that attempt to interview and identify either the inmovers or the outmovers since Census Day. This involves using name and demographic characteristics such as age, sex, and date of birth to match the person records. When matching, the survey-based approach typically uses a search area definition of a small area where the person was supposed to be enumerated on Census Day. In our application, the EM log-linear model uses the Protected Identification Key assigned to each record for matching. With this approach, we implemented Definition 2 that uses a national search area for matching.

Poststratification and heterogeneous independence involve a third set of assumptions. The main assumption is that the population can be divided into groups within which the capture probabilities are homogenous for at least one of the lists. Starting in the 2010 Census, the survey-based approach used logistic regression models to do this[12], which is comparable to the loglinear modeling of the AR approach. The large amount of data used in the AR modeling reduces the statistical uncertainty in the estimates of the model parameters compared to that which arises from the PES modeling being based on a sample. For Mth, the poststratification assumption is that every characteristic or domain is accurately recorded for the census and the survey, so that people are assigned to the correct poststratum. The EM log-linear method in our study facilitates dealing with

[12] For logistic regression in the PESs, all the regression variables were categorical, which resulted in a finite, though large, set of cross-classified categories with nonzero cells for the models. Every person within any specific one of these cells would get the same estimated probabilities of correct enumeration or census inclusion from the model.

characteristics that are collected with differences by the census and administrative data in our second system.

Another assumption related to the homogeneity assumption about capture probabilities, is that each person should have a positive probability of being captured in each system. When using a series of administrative data sources for the second system there is the possibility that some parts of the population could be systematically missed, not just randomly missed, by all the systems. If there is enough of a population that is impacted by this violation of the homogeneity assumption, then that can lead to bias in the estimates.

Use of Sampling

The usage of a sample has advantages and disadvantages. The PES uses a sample while the administrative record-based approach in our study does not. Since the PES uses a sample, one advantage is that it is representative of the entire household population. The sampled people can be given attention not possible for the much larger AR data system. This includes augmenting computer matching with clerical matching and follow-up to reduce false matches and false nonmatches. A disadvantage is that the resulting sampling error may mean that direct population estimates would be reliable only for areas or population groups with reasonably large representation in the sample. This can be a particular problem for population estimates for small race groups. Modeling with sample data usually requires that the modeling account for the sample design, which can complicate the model fitting and determining uncertainty measures. The AR approach avoids these disadvantages and complications since it uses full administrative data systems. However, since the administrative sources not produced to be representative of the full population, there is the possibility of some segments of the population being systematically excluded, leading to bias. Combining multiple AR sources can also produce duplicate records that go undetected. Finally, it must be noted that synthetic error in coverage estimates is a limitation for both the PES and AR approaches.

Relevance and Quality of the Data

Relevance and quality of the data obtained is another area where there are advantages and disadvantages for AR and survey-based approaches. Survey-based approaches have the advantages of conducting interviews that are designed to get exactly the information needed about the sample cases to facilitate the needed usages in DSE. Intensive follow-up can be done as needed to improve the overall data quality or deal with issues that surface. Each PES can incorporate improvements to the questionnaire, field procedures and other

operational aspects. There are two disadvantages of survey-based approaches. One is the need to at least attempt to minimize dependence between the census and PES data collections that results in the PES interviewing in the United States being delayed until well after Census Day. This delay can create recall problems for respondents. Also, some people will move between the census and the PES interview day, so special procedures have to be developed to account for these movers.

Administrative data have the advantage of possibly containing high quality information on certain aspects of the population (e.g., birth dates instead of reported ages) especially when multiple sources are used. The AR approach also has the potential advantage of minimizing the dependence between the census and AR to the extent that they are compiled independently. However, the use of AR in the Census enumeration, as in imputation in the 2020 census, could compromise independence. A disadvantage of the AR approach is that while each AR source may provide high quality information related to the purpose of administering a certain program, this information may differ from the information needed for the estimation of the population on April 1. Some information, such as demographic characteristics, may not be available from some AR sources. Also, if an administrative program changes, then the usefulness of its data for our purposes may be impacted. Programs may collect information like race or Hispanic origin differently than our census questionnaire. While we can attempt to use multiple sources to potentially mitigate these disadvantages, we do not have direct control over this input source like we do with the PES interview.

Matching Implications Especially Related to the Search Area Used

Matching implications, especially in relation to the search area used, is another aspect with advantages and disadvantages for both approaches. The PES implements matching of people within a small geographic search area. The survey-based approach done by the Census Bureau is a one-way match from the independent interview to census enumerations that combines computer and clerical matching. The use of the two types of matching conducted in a small geographic area leads to a reduction in false matches. The small geographic area provides a manageable number of correct census records to be examined to determine if the record is a match or nonmatch. The onus of this approach is to determine the small geographic area where the person should be counted in the census. This can be challenging for how the survey-based approach handles movers between census day and the survey interview date. The survey-based approach implemented in 2010 and 2020 had to gather information during the interview to determine the small geographic area to search for the census enumeration.

Our implementation of the administrative record approach has conducted nationwide matching. To minimize false matches, our approach has used a stricter criterion of sufficient name and characteristic information based on the census and administrative records having a PIK assigned. The matching between the two systems is done based on records having the same PIK or not. While this facilitates large scale matching and the stricter criterion for matching helps to minimize matching error, the extent to which the exclusion of some people who do not get a PIK but are true residents results in more omissions that need to be accounted for by the DSE. While this approach may have advantages for national-level coverage estimates, a disadvantage for subnational estimates is that the census and the AR system may have some people in different geographic areas, leading to uncertainty about where such people should be counted.

4. Results

This section highlights the net coverage error estimates generated by our study. In this project, we generated estimates for various age groups. Where applicable, our net coverage error estimates are compared to the net coverage error estimates generated by the DA and the PEP. Positive estimates indicate an overcount and negative estimates indicate an undercount. In section 4.1, we show results for the 0-to-4 age grouping. In Section 4.2, we show the results for the 30- to 49-year-old age grouping.

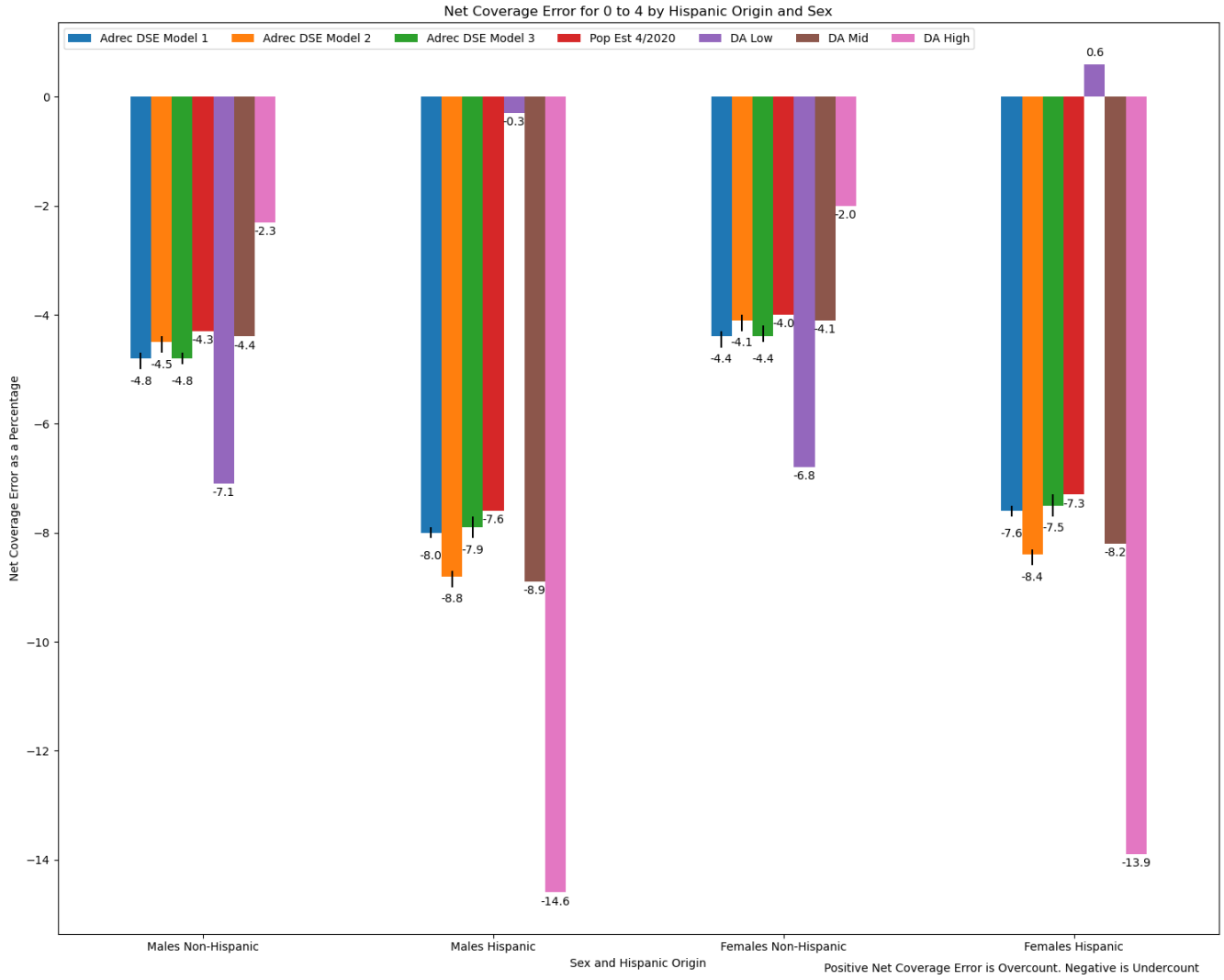
4.1 0-to-4 Age Group Results

This section highlights some of the results and comparison for the 0-to-4 age group. Figure 1 shows the net coverage error results for April 1, 2020, for the 0-to-4 population by Hispanic origin. Results from three EM log-linear models are compared to the net coverage error results from the PEP and DA.

- For non-Hispanic males, our study estimated net coverage error ranging from -4.5 to -4.8 percent. The PEP and DA middle series had similar estimates of -4.3 and -4.4 percent, respectively. DA had a low to high range of -2.3 to -7.1 percent.
- For non-Hispanic females, our study estimated net coverage error ranging from -4.1 to -4.4 percent. The PEP and DA middle series had similar estimates of -4.0 and -4.1 percent, respectively. DA had a low to high range of -2.0 to -6.8 percent.
- For Hispanic males, our study estimated net coverage error ranging from -7.9 to -8.8 percent. The PEP and DA middle series had similar estimates of -7.6 and -8.9 percent, respectively. DA had a low to high range of -0.3 to -14.6 percent.

- For Hispanic females, our study estimated net coverage error ranging from -7.5 to -8.4 percent. The PEP and DA middle series had similar -7.3 and -8.2 percent, respectively. DA had a low to high range of 0.6 to -13.9 percent.

Figure 1: Net Coverage Error for 0-to-4 Population by Hispanic Origin



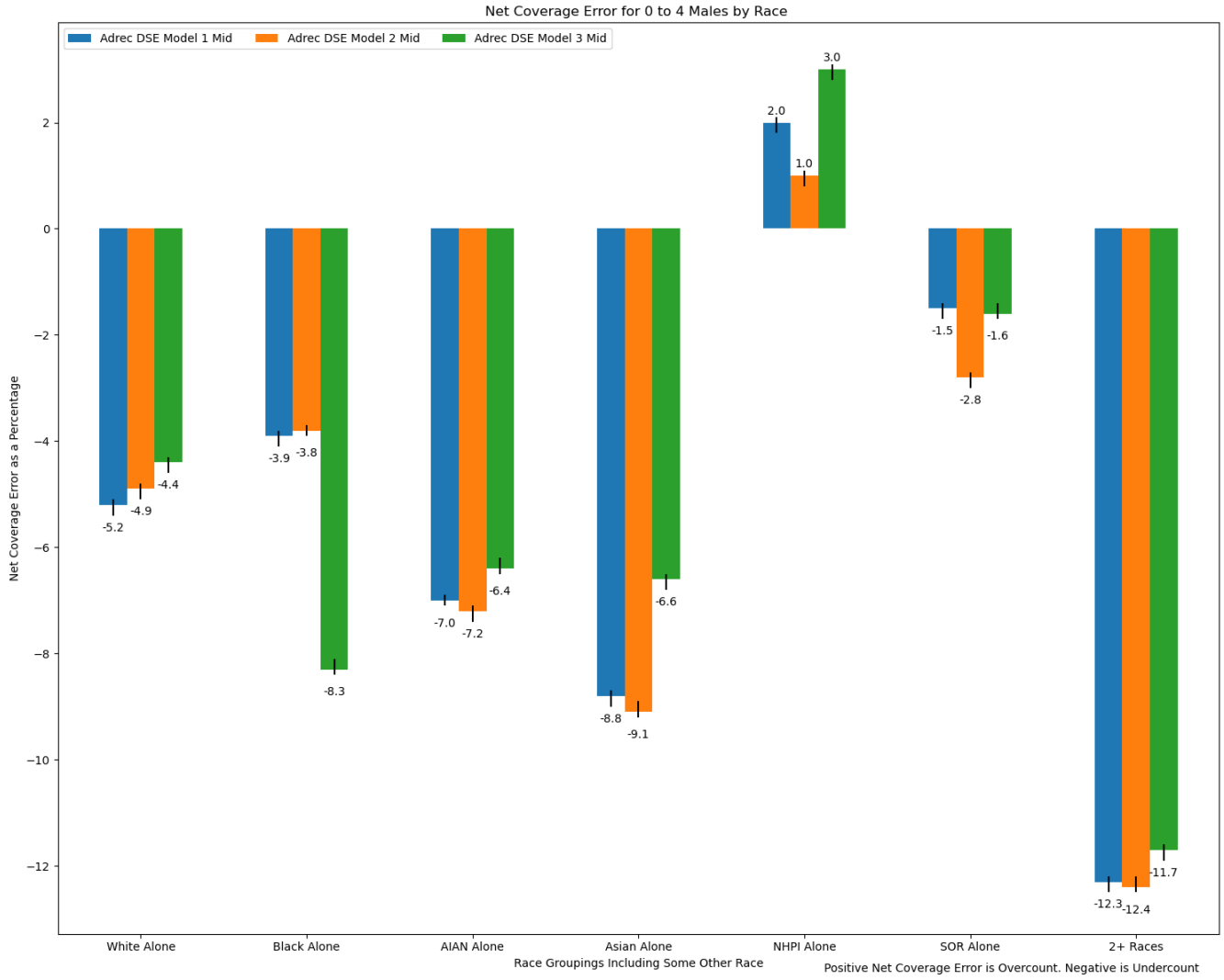
Source: U.S. Census Bureau, 2020 Census Edited File, 2020 Demographic Analysis, 2020 Population Estimates Program and 2020 Adrec DSE

Figure 2 and Figure 3 shows the net coverage error results for April 1, 2020, for the 0-to-4 males and females, respectively, by race groups that include some other race. Results from three EM log-linear models are shown. There are no comparable DA or PEP results.

- For the White alone population, our study estimated net coverage error between -4.4 and -5.2 percent for males. Our study estimated net coverage error between -4.0 and -4.8 percent for females.

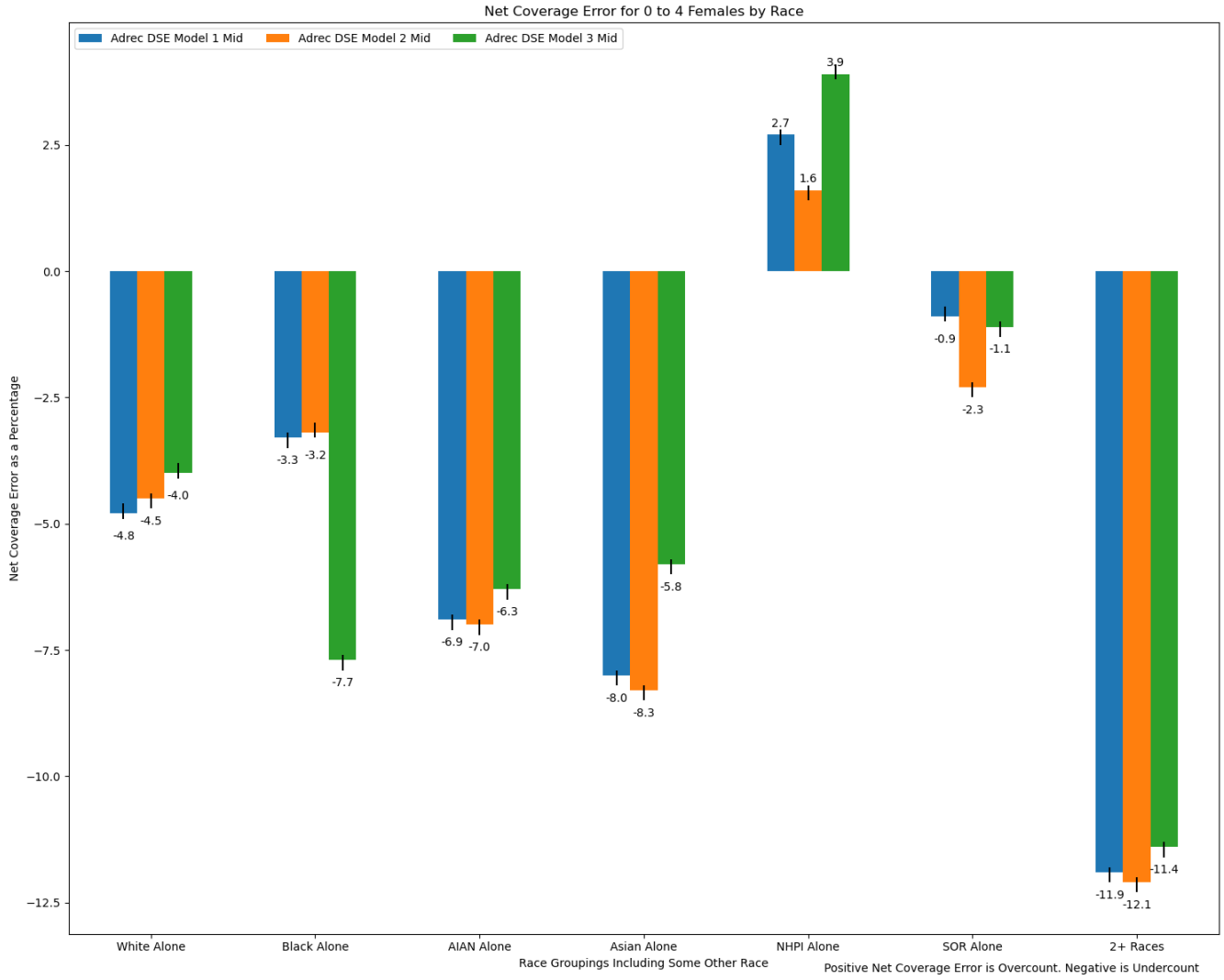
- For the Black alone population, our study estimated net coverage error between -3.8 and -8.3 percent for males. Our study estimated net coverage error between -3.2 and -7.7 percent for females.
- For the AIAN alone population, our study estimated net coverage error between -6.4 and -7.2 percent for males. Our study estimated -6.3 and -7.0 percent for females.
- For the Asian alone population, our study estimated net coverage error between -6.6 and -9.1 percent for males. Our study estimated -5.8 and -8.3 percent for females.
- For the NHPI alone population, our study estimated net coverage error between 1.0 and 3.0 percent for males. Our study estimated 1.6 to 3.9 percent for females.
- For the Some Other Race Alone population, our study estimated net coverage error between -1.5 and -2.8 percent for males. Our study estimated -0.9 to -2.3 percent for females.
- For the multiracial population, our study estimated net coverage error between -11.7 and -12.4 percent for males. Our study estimated between -11.4 and -12.1 for females.

Figure 2: Net Coverage Error for 0-to-4 Males by Race Including Some Other Race



Source: U.S. Census Bureau, 2020 Census Edited File and 2020 Adrec DSE

Figure 3: Net Coverage Error for 0-to-4 Females by Race Including Some Other Race



Source: U.S. Census Bureau, 2020 Census Edited File and 2020 Adrec DSE

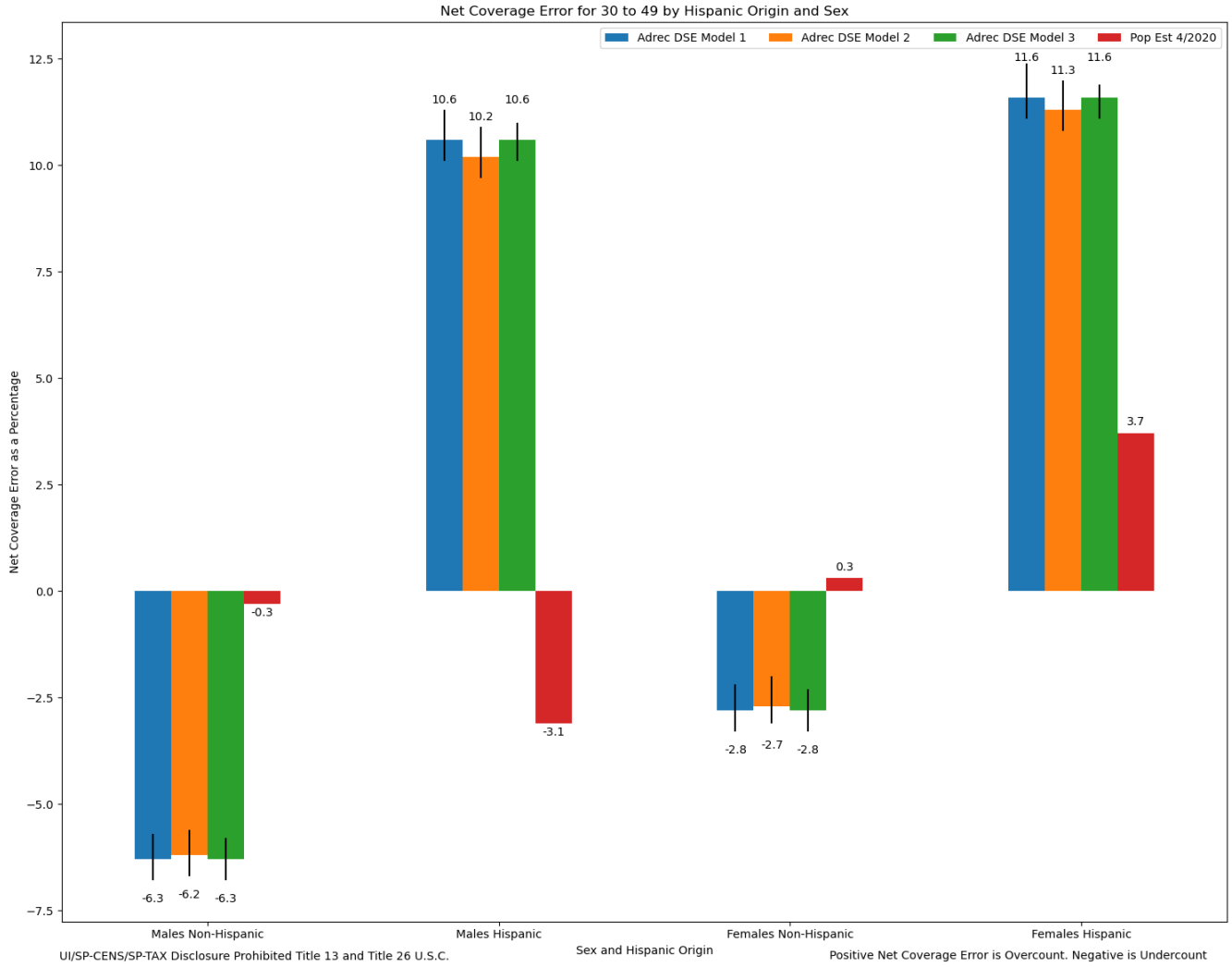
4.2 30-to-49 Age Group Results

This section shares results and comparisons for the 30- to 49-year-old age group. Figure 4 shows the net coverage error results for April 1, 2020, for the 30-to-49 population by Hispanic origin. Results from three EM log-linear models are compared to the net coverage error results from the PEP. The December 2020 release of the DA estimates only did Hispanics up to age 29.

- For non-Hispanic males, our study estimated net coverage error ranging from -6.2 to -6.3 percent. The PEP has a smaller undercount estimate of -0.3 percent.

- For non-Hispanic females, our study estimated net coverage error ranging from -2.7 to -2.8 percent. The PEP has an overcount estimate of 0.3 percent.
- For Hispanic males, our study estimated net coverage error ranging from 10.2 to 10.6 percent. The PEP has an undercount estimate of -3.1 percent.
- For Hispanic females, our study estimated net coverage error ranging from 11.3 to 11.6 percent. The PEP also has an overcount estimate of 3.7 percent.

Figure 4: Net Coverage Error for 30-to-49 Population by Hispanic Origin



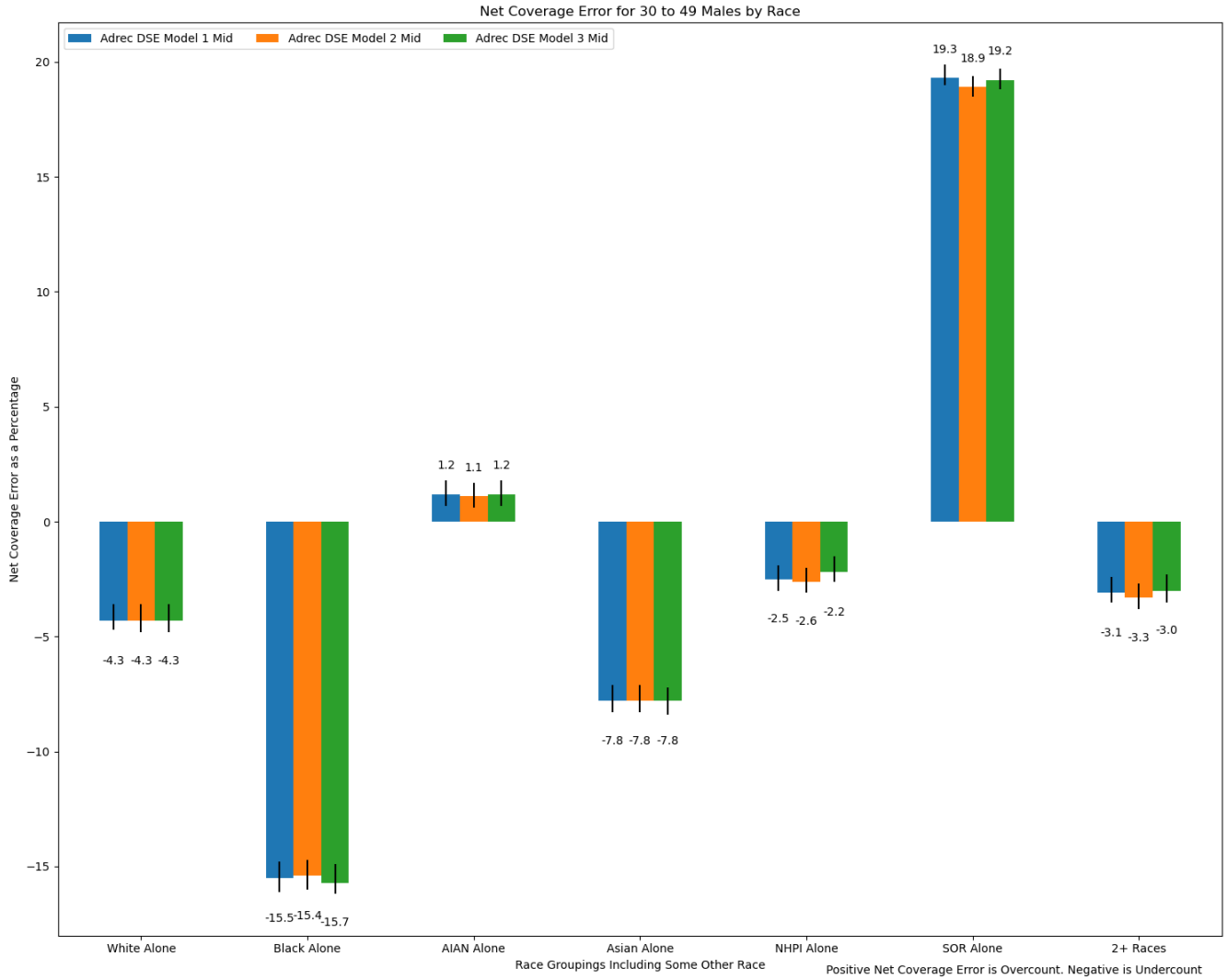
Source: U.S. Census Bureau, 2020 Census Edited File, 2020 Population Estimates Program and 2020 Adrec DSE

Figure 5 and Figure 6 show the net coverage error results for April 1, 2020, for the 30-to-49 males and females, respectively, by race groups that include some other race.

Results from three EM log-linear models are shown. There are no comparable DA or PEP results.

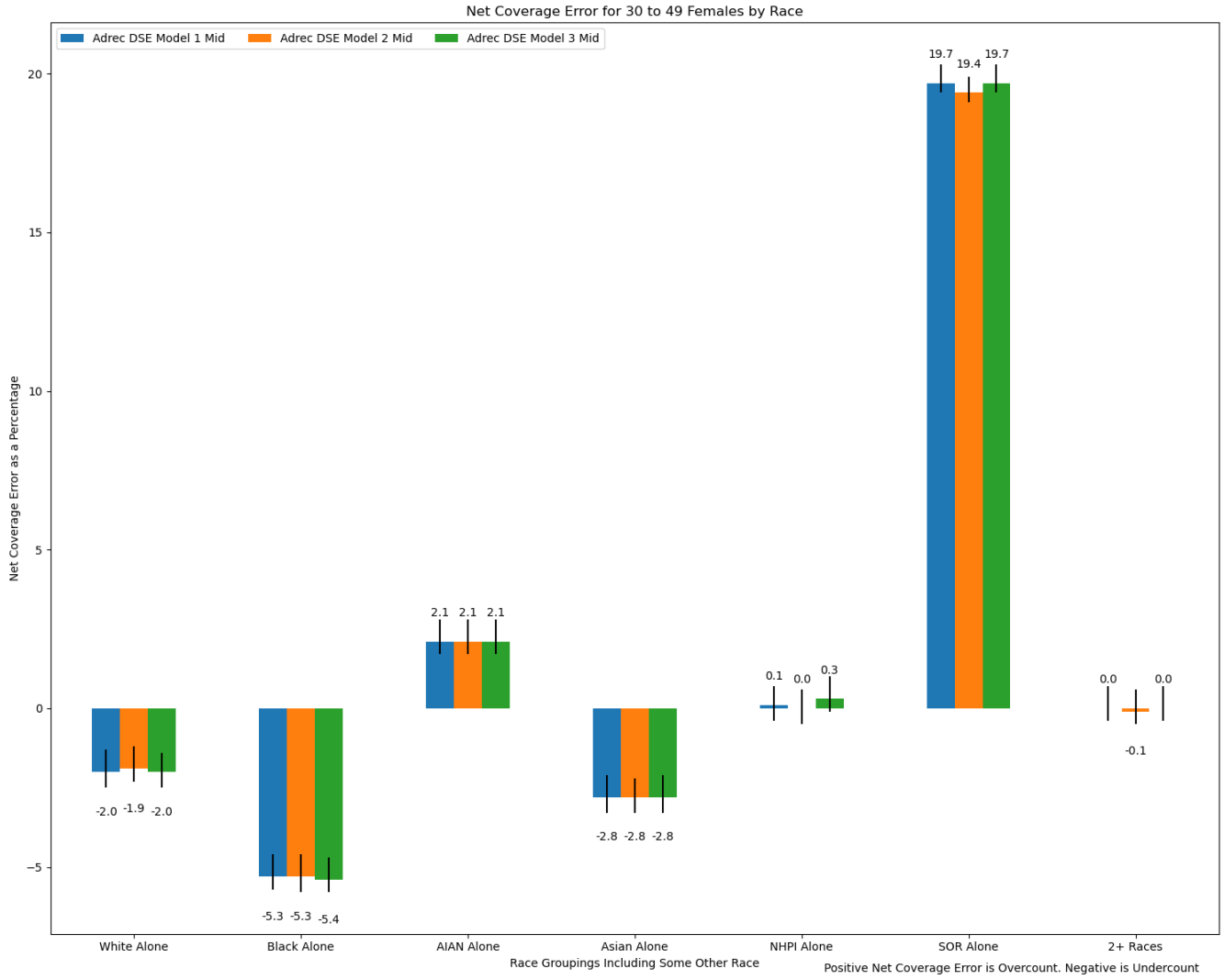
- For the White alone population, our study estimated net coverage error of about -4.3 percent for males for all three models. Our study estimated net coverage error between -1.9 and -2.0 percent from all three models for females.
- For the Black alone population, our study estimated net coverage error between -15.4 and -15.7 percent for males. Our study estimated net coverage error between -5.3 and -5.4 percent for females.
- For the AIAN alone population, our study estimated net coverage error between 1.1 and 1.2 percent for males. Our study estimated net coverage error of 2.1 percent from each of the three models for females.
- For the Asian alone population, our study estimated net coverage error of -7.8 percent from the three models for males. Our study estimated -2.8 percent from the three models for females.
- For the NHPI alone population, our study estimated net coverage error between -2.2 and -2.6 percent for males. Our study estimated net coverage error between 0.0 to 0.3 percent for females. The confidence intervals for females covered zero.
- For the Some Other Race Alone population, our study estimated net coverage error between 18.9 and 19.3 percent for males. Our study estimated net coverage error between 19.4 to 19.7 percent for females.
- For the multiracial population, our study estimated net coverage error between -3.0 and -3.3 percent for males. Our study estimated net coverage error between -0.1 and 0.0 for females. The confidence intervals for females include 0 percent.

Figure 5: Net Coverage Error for 30-to-49 Males by Race Including Some Other Race



Source: U.S. Census Bureau, 2020 Census Edited File and 2020 Adrec DSE

Figure 6: Net Coverage Error for 30-to-49 Females by Race Including Some Other Race



Source: U.S. Census Bureau, 2020 Census Edited File and 2020 Adrec DSE

5. Conclusions

Mainly, this project was able to successfully produce capture-recapture population estimates that used administrative data for the second system as compared to data from a post-enumeration survey. The project identified and modified an approach that was implemented by other national statistical agencies and applied it to a possible application for the decennial census. This work focused on generating estimates when race and Hispanic origin information might be recorded differently in the 2020 Census enumeration versus the information available from a past census or from administrative data. This may be also beneficial since the Office of Management and Budget revised Statistical Policy Directive Number 15 for collecting federal data on race and ethnicity. While the results for

race and Hispanic origin being recorded differently were a focus of this report, this approach can be applied to other instances like state of residence.

This project produced estimates of net census coverage for racial and Hispanic origin groups by age and sex. Such detailed estimates were not available in the initial March 2022 release of results from the Post-Enumeration Survey, but such results were eventually released by the program. The AR DSEs use a second system with a huge amount of data compared to the much smaller sample sizes of the Post-Enumeration Survey, which leads to significant sampling error in estimates for some detailed cross-classified demographic groups.

This report presents results of population estimates and corresponding estimates of net census coverage based on three different models. Two of these models use additional covariates related to the Hispanic or non-Hispanic Black alone concentration of tracts based on ACS 5-year estimates. Our results show that these models can, in some instances, produce different population estimates especially for the Hispanic and Black populations. Our report shows all three estimates reflect uncertainty. Additional work could be done to develop methods that can produce appropriate uncertainty measures for this type of analysis.

This report also shows instances like estimates for 30- to 49-year-old Hispanics where additional work is needed. Our research shows an overcount for this Hispanic population which was not seen in the PEP. This is a group where possibly additional data sources, like for the group quarters populations, may be needed. The work in this report only used two data systems. This may be an area where the implementation of multiple system estimation may be beneficial. For our application, the American Community Survey may be a candidate for a third system that would not require doing a separate post-enumeration survey operation.

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Appendix A: Log-Linear Models

<table>
<tr><td>National</td><td>ACS 5-year Hispanic</td><td>ACS 5-year NH Black Alone</td></tr>
<tr><td colspan="3">Census system (a)
Adrec system (b)
Census race/Hispanic origin (x1)
Adrec race/Hispanic origin (x2)
Census system x Adrec race/Hispanic origin
Adrec system x Census race/Hispanic origin
Census race/Hispanic origin x Adrec race/Hispanic origin
2010 PIK Covariate
2010 PIK Covariate x Census race/Hispanic origin
2010 PIK Covariate x Adrec race/Hispanic origin
2010 PIK Covariate x Census system
2010 PIK Covariate x Adrec System
2010 PIK Covariate x Census system x Adrec race/Hispanic origin
2010 PIK Covariate x Adrec system x Census race/Hispanic origin
2010 PIK Covariate x Census race/Hispanic origin x Adrec race/Hispanic origin</td></tr>
<tr><td>No additional covariates</td><td>ACS 5-year Hispanic Grouping
<10%
10% to <50%
50%+
ACS Group x Census race/Hispanic origin
ACS Group x Adrec race/Hispanic origin
ACS Group x Census system
ACS Group x Adrec System
ACS Group x Census system x Adrec race/Hispanic origin
ACS Group x Adrec system x Census race/Hispanic origin
ACS Group x Census race/Hispanic origin x Adrec race/Hispanic origin</td><td>ACS 5-year NH Black Alone Grouping
<10%
10% to <50%
50%+
ACS Group x Census race/Hispanic origin
ACS Group x Adrec race/Hispanic origin
ACS Group x Census system
ACS Group x Adrec System
ACS Group x Census system x Adrec race/Hispanic origin
ACS Group x Adrec system x Census race/Hispanic origin
ACS Group x Census race/Hispanic origin x Adrec race/Hispanic origin</td></tr>
</table>